

A Healthy Approach

A course update to encompass KEEN outcomes transforms a biomedical engineering program



BME students observe as Mansoor Nasir (right) and Eric Meyer demonstrate how data from the accelerometer, gyroscope, and pressure sensors in the Nike+ Shoe can be used to monitor and improve user performance.

Go to any gym across America, and you will likely see evidence of “quantified self” (QS) technology. Fitness buffs are wearing devices that monitor heart rate, calorie burn, speed, and other data bits that enhance awareness and performance. Technology once limited to intensive care units in elite hospitals is now ubiquitous. The application of QS technology has become a social movement where people use biometric sensors in smartphones and wearable devices to track and analyze personal health data ranging from steps taken to sleep cycles.

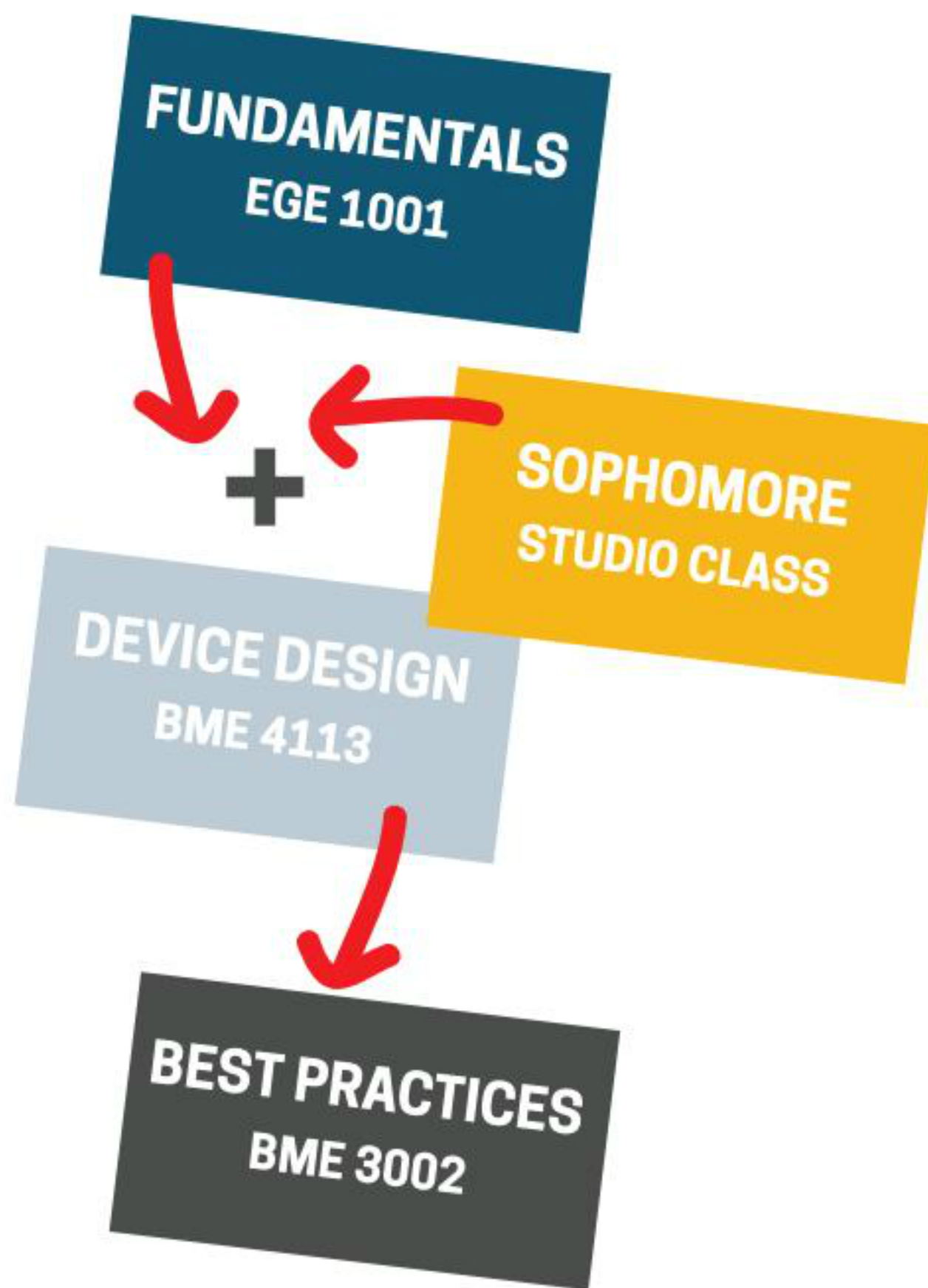
Eric Meyer and Mansoor Nasir, both assistant professors of biomedical engineering at Lawrence Technological University in Southfield, Mich., saw the QS craze as an ideal way to teach entrepreneurial mindedness to first-year biomedical students.

“We were trying to find a way to present students with engaging, accessible projects in their

earliest courses so that our students would develop confidence in approaching open-ended problems,” says Meyer.

Through a KEEN topical grant, Meyer and Nasir were able to teach an early biomedical engineering course from a new entrepreneurial perspective.

“Typically, biomedical engineering courses focus on medical devices, which is an area of engineering that is difficult to work in due to the stringent regulatory and marketing environment of the U.S. healthcare industry,” Meyer says. “But the emerging fitness device industry in many cases closely aligns with similar medical devices, only with a much more entrepreneurial spirit and much less regulation. We saw QS as a timely, exciting, real-world entrepreneurship opportunity for our students to work on using the technical skills they were learning in courses across the curriculum.”



Because QS combines miniaturization, biomedical testing technology, social media, and interest in personal improvement, it is ideal for applying active, collaborative, interdisciplinary, and problem-based pedagogical techniques.

"Teaching QS requires technical instruction in human physiology and sensors, computer science and software development, networking, data analysis, and visualization," says Meyer. "But it also provides rich discussion opportunities for non-technical issues such as privacy, medical ethics, regulations, intellectual property, social issues, and marketing."

The field is highly accessible to students – many carry QS technology in their pockets, Nasir notes. That familiarity makes it easier for students to practice the market-based skill of opportunity recognition. Building on this foundation will help students foster the entrepreneurial skills necessary for more advanced projects.

Meyer and Nasir initially intended to modify one or two courses to instill the entrepreneurial mindset. "But then we realized the modules could

be incorporated throughout the course sequence, even from the earliest introductory course, to ramp up students' entrepreneurial skills year-by-year in preparation for the meaty projects of their later courses," Meyer says.

The movement from a simple course upgrade to a program-wide application started with a conversation among biomedical engineering faculty when preparing for ABET accreditation. The staff realized the focus of their program favored narrow research at the expense of design – so much so that some students' first foray into open-ended engineering was in their senior capstone course.

"It's sort of unfair to expose students to that for the first time in their senior project," says Nasir.

"We decided we needed to do more to promote engineering design and give our students opportunities throughout the program to gain experience and broaden their thinking," Meyer adds. "Our experience showed us that most of our students are initially uncomfortable with assignments and projects that are completely open-ended. If we can help by giving them a target and some familiar, real-world examples, some of their trepidation can be alleviated."

When handed open-ended projects, students tend to design a device or system from scratch, says Nasir. If they find their idea has flaws or that others are working on a similar concept, they view it as a failure. "But in the real world this happens all the time," he says. "Engineers revisit old devices, repeat what others in the industry are working on, and sometimes find their idea doesn't work as hoped. These are all steps in developing better products.

"The first thing you have to get students to realize is that in the real world, problems don't present themselves where 'here is the problem, here is the answer,'" he adds. "In many cases the problems aren't obvious, and the answers are many."

Now the entrepreneurial approach is being woven throughout the biomedical engineering program. "We have embraced the KEEN belief that entrepreneurship is a mindset and that the entrepreneurial process can be formalized," Meyer says. "We are modifying courses across the curriculum to train students to stop thinking only like an engineer or scientist and start thinking like a product developer."

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Entrepreneurial engineering opportunities in QS abound due to accelerator and developer programs. Last year, investors pumped hundreds of millions of dollars into digital medical devices and wearable biosensors. This excites faculty and students alike, from business opportunity and academic perspectives, since market dynamics are changing rapidly and students are familiar with the products.

Nasir notes that this could not have happened without collaboration that The Kern Family Foundation encourages. As the program matures, the LTU biomedical engineering team aims to work with other departments at LTU and even other universities within KEEN.

"By collaborating with faculty from many engineering and life science disciplines, we could distribute these experiences broadly across courses and at various levels of the curriculum," Meyer says. "So far we have five faculty from various departments and KEEN universities interested in building a resource base for pedagogical techniques promoting entrepreneurial engineering mindsets. We're excited to work together to develop more modules, and we encourage others to consider using these modules across biomedical engineering programs."



Nasir demonstrates the iHealth wireless blood pressure wrist monitor to BME students. Data is recorded on the app allowing users to track personal health trends.

BUILDING ENTREPRENEURIAL SKILLS INTO BIOMEDICAL CURRICULUM

Using project-based learning modules developed through KEEN Topical Grant funding, professors at Lawrence Technological University have made the following additions into four of their courses:

■ INTRO TO BIOMEDICAL ENGINEERING (FRESHMAN)

Using the Nike+ Sensor Suite to practice a "painstorming" process, develop new devices, conduct market research, and present business plans that culminate in elevator speeches.

■ BME BEST PRACTICES (JUNIOR)

"Repurposing" QS devices for Global Health needs and navigating the intellectual property regulations involved.

■ MEDICAL DEVICE DESIGN (JUNIOR)

Design prototype or model to address student-identified opportunities and present customer and economic value.

■ ORTHOPEDICS (SENIOR)

Individual research into technical issue in orthopedics, identifying opportunity, reviewing recently approved relevant devices and patents, and submitting manual for Good Manufacturing Practice.